

Conservation of Energy Solutions

Question 1 (Conceptual)

The work–energy theorem, in the context of conservative forces (like gravity), states that the total mechanical energy of a system is conserved if no non-conservative forces (like friction or air resistance) do work on the system.

Total Mechanical Energy (E):

$$E = \text{Kinetic Energy}(K) + \text{Potential Energy}(U_g)$$

$$E = \frac{1}{2}mv^2 + mgh$$

Since friction is neglected, gravity is the only force doing work. Gravity is a conservative force.

Comparison: The roller coaster car's total mechanical energy **remains the same** (is conserved) at both the top of the track and the point 30 m below. At the top, the energy is purely potential. As the car moves down, the potential energy (U_g) decreases, and the kinetic energy (K) increases, but their sum (E) stays constant.

Question 2 (Calculation)

Given values:

- Mass of the block (m): 2.0 kg
- Initial height (h): 10 m

- Length of the incline (L): 20 m
- Gravitational acceleration (g): 9.8 m/s^2

(a) Identify the equation for conservation of mechanical energy.

Since the incline is initially frictionless, the total mechanical energy is conserved:

$$E_{\text{initial}} = E_{\text{final}}$$

$$K_i + U_{g,i} = K_f + U_{g,f}$$

$$\frac{1}{2}mv_i^2 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f$$

(b) Calculate the block's speed at the bottom.

- Initial speed (v_i): 0 m/s (released from rest)
- Final height (h_f): 0 m (at the bottom)
- Initial height (h_i): 10 m

The conservation equation simplifies to:

$$0 + mgh_i = \frac{1}{2}mv_f^2 + 0$$

The mass (m) cancels out:

$$gh_i = \frac{1}{2}v_f^2$$

Solve for the final speed (v_f): $v_f = \sqrt{2gh_i}$

Substitute the values:

$$v_f = \sqrt{2(9.8 \text{ m/s}^2)(10 \text{ m})}$$

$$v_f = \sqrt{196 \text{ m}^2/\text{s}^2}$$

$$v_f = 14 \text{ m/s}$$

(c) If a constant friction force of 15 N acts along the incline, how much mechanical energy is lost to heat?

The energy lost is equal to the work done by the non-conservative friction force (W_{nc}), which is negative.

$$W_{nc} = \vec{F}_{friction} \cdot \vec{d}$$

The friction force (F_f) is 15 N and acts opposite to the displacement (L). The displacement is 20 m.

$$W_{friction} = -F_f L$$

$$W_{friction} = -(15 \text{ N})(20 \text{ m})$$

$$W_{friction} = -300 \text{ J}$$

The amount of mechanical energy lost to heat is the magnitude of the work done by friction: 300 J.

(d) What is the block's speed at the bottom with friction?

The full work–energy theorem states that the work done by non-conservative forces equals the change in total mechanical energy (ΔE):

$$W_{friction} = \Delta E = E_{final} - E_{initial}$$

$$W_{friction} = \left(\frac{1}{2}mv_f^2 + mgh_f \right) - \left(\frac{1}{2}mv_i^2 + mgh_i \right)$$

Substitute $v_i = 0$ and $h_f = 0$:

$$W_{friction} = \frac{1}{2}mv_f^2 - mgh_i$$

Rearrange to solve for v_f :

$$\frac{1}{2}mv_f^2 = mgh_i + W_{friction}$$

$$v_f^2 = \frac{2}{m}(mgh_i + W_{friction})$$

$$v_f = \sqrt{\frac{2}{m}(mgh_i + W_{friction})}$$

We know: $mgh_i = (2.0 \text{ kg})(9.8 \text{ m/s}^2)(10 \text{ m}) = 196 \text{ J}$.

$$v_f = \sqrt{\frac{2}{2.0 \text{ kg}}(196 \text{ J} - 300 \text{ J})}$$

We find the work of friction is $W_{friction} = -300 \text{ J}$. Since the total initial energy is only 196 J, the friction work of 300 J is greater than the total available energy.

Thus, the block can only travel a distance ΔL where the work of friction equals the total initial potential energy, mgh .

$$F_f \Delta L = mgh$$

$$\Delta L = \frac{196 \text{ J}}{15 \text{ N}} \approx 13.07 \text{ m}$$

Since the incline is 20 m long, the block stops prior to reaching the bottom. The total initial energy is 196 J. The energy lost is 300 J. The block comes to a stop when its kinetic energy is zero, which means its final speed is 0 m/s.

As the block does not reach the end of the incline, its speed at the end of the incline does not exist.